Nemanja Mikić

Senior Software Engineer · Backend, Platform & Cloud · Java, Python, Go

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Professional Summary

Software engineer with 8+ years of experience building backend services, developer frameworks and cloud integrations in Java, Python and Go. At Oracle since 2022 (IC4): 400+ merged pull requests across the Micronaut ecosystem, including OpenTelemetry tracing, vector search in Micronaut Data, Oracle Cloud (OCI) and AWS integrations, Kubernetes workload identity and GraalPy interoperability. Author of official Micronaut Guides; credited in Apple's web server security acknowledgements (November 2025).

Core Skills

Languages: Java, Python, Go, SQL

Backend: Micronaut, Netty, GraalVM native image, GraalPy, REST and gRPC APIs, microservices, JDBC, R2DBC, Hibernate

Cloud and DevOps: Oracle Cloud (OCI), AWS, Google Cloud, Kubernetes, Docker, Terraform, CI/CD (GitHub Actions, GitLab CI), Gradle, Maven

Observability and data: OpenTelemetry, Micrometer, Prometheus, CloudWatch, PostgreSQL, MySQL, Oracle Database, MongoDB, vector search

Security: Application security, penetration testing, OAuth2, OpenID Connect, SSO, MFA/TOTP, HashiCorp Vault, TLS certificates

Experience

Oracle — Software Engineer (IC4)

Jan 2022 – Present

IC4 since June 2024; IC3 from January 2022 to May 2024 · 400+ merged pull requests across the Micronaut ecosystem · Belgrade

- Designed and shipped Micronaut's OpenTelemetry integration (HTTP, gRPC, JDBC, R2DBC) and the Micrometer Observation and Prometheus PushGateway modules, merged into the official micronaut-projects repositories with tests and documentation.
- Delivered Micronaut Data vector search end to end: dense and sparse vector types, derived similarity queries, schema and index generation, and JDBC/R2DBC bindings for Oracle Database, PostgreSQL (pgvector) and MySQL HeatWave.
- Built the OCI integrations for logging, certificate management and OKE Kubernetes workload identity, and the AWS CloudWatch logging module; fixed OCI SDK retry back-off that could block Netty event loops and stall server I/O.
- Extended an internal Java service framework with metrics, HTTP/JVM instrumentation and request-ID propagation, and wrote its Terraform reference deployments.
- Advanced Python on Micronaut (Pyronaut, GraalPy): coroutine-to-Publisher adaptation, automatic dataclass introspection with validation, pytest support, PyPI releases and native-image packaging.
- Built Graal Development Kit application and Kubernetes generation for AWS, OCI and GCP, and GitLab CI tooling for automated dependency updates, releases, smoke tests and OCI authentication.
- Authored five official Micronaut Guides and co-authored a sixth (Kubernetes on OKE, EKS and GKE; metrics on CloudWatch and OCI; OpenTelemetry tracing on OCI), each with a runnable, tested application.

Seven Bridges Genomics — Penetration Tester

May 2021 – Dec 2021

- Moved to the security team to test the platform I had helped build: web and API penetration testing, authentication and authorization reviews, and remediation guidance for the Core Team.

Seven Bridges Genomics — Software Engineer, Core Team

Dec 2017 – May 2021

- Built platform services in Java, Python and Go, including shared secrets management with HashiCorp Vault and MFA with recovery codes, email verification and TOTP.
- Split a monolithic billing service into separate charging and pricing services in Java, improved OAuth2 single sign-on, and built the administrative billing API gateway and Go HTTP clients.
- Automated shutdown of idle AWS EC2 test environments and built a notification service for targeted and platform-wide messages (Go, MongoDB).

Seven Bridges Genomics — Python Software Engineer Intern

Mar 2017 – Jun 2017

- Built a Python user management service for registration and authentication, plus a frontend API gateway with Tornado, SQLAlchemy and PostgreSQL.

Education

MSc in Software Engineering — University of Belgrade, School of Electrical Engineering — Oct 2017 – Sep 2019

BSc in Software Engineering — University of Belgrade, School of Electrical Engineering — Oct 2013 – Sep 2017

Languages: Serbian (native), English (professional working proficiency)